

Strong and Weak Forms of Solid Mechanics: Spatial, Referential, and Total Lagrangian Formulations

1 Kinematics and Notation

Let $\Omega_R \subset \mathbb{R}^d$ denote the reference (material) configuration of a body (referential quantities carry the subscript R) with boundary $\partial\Omega_R = \Gamma_R^D \cup \Gamma_R^N$, $\Gamma_R^D \cap \Gamma_R^N = \emptyset$, and let Ω_t denote the current (spatial) configuration at time t with boundary $\partial\Omega_t = \Gamma_t^D \cup \Gamma_t^N$. The motion

$$\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t), \quad \mathbf{X} \in \Omega_R, \quad (1)$$

maps material points $\mathbf{X}$ to spatial points $\mathbf{x}$. The displacement, deformation gradient, and Jacobian are

$$\mathbf{u}(\mathbf{X}, t) = \boldsymbol{\varphi}(\mathbf{X}, t) - \mathbf{X}, \quad \mathbf{F} = \text{Grad } \boldsymbol{\varphi} = \mathbf{I} + \text{Grad } \mathbf{u}, \quad J = \det \mathbf{F} > 0, \quad (2)$$

where $\text{Grad}(\cdot) = \partial(\cdot)/\partial\mathbf{X}$ and $\text{grad}(\cdot) = \partial(\cdot)/\partial\mathbf{x}$ denote referential and spatial gradients, respectively. The right Cauchy–Green tensor and the Green–Lagrange strain are

$$\mathbf{C} = \mathbf{F}^\top \mathbf{F}, \quad \mathbf{E} = \frac{1}{2} (\mathbf{C} - \mathbf{I}) = \frac{1}{2} \left(\text{Grad } \mathbf{u} + \text{Grad}^\top \mathbf{u} + \text{Grad}^\top \mathbf{u} \text{ Grad } \mathbf{u} \right). \quad (3)$$

The velocity is $\mathbf{v} = \dot{\boldsymbol{\varphi}} = \dot{\mathbf{u}}$ (material time derivative). Mass conservation relates the referential and spatial densities by $\rho_R = J\rho$.

The stress measures used below are the Cauchy stress $\boldsymbol{\sigma}$ (force per unit *current* area), the first Piola–Kirchhoff stress $\mathbf{P}$ (force per unit *reference* area), and the second Piola–Kirchhoff stress $\mathbf{S}$, related through the Piola transform

$$\mathbf{P} = J\boldsymbol{\sigma}\mathbf{F}^{-\top}, \quad \mathbf{S} = \mathbf{F}^{-1}\mathbf{P} = J\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-\top}, \quad \boldsymbol{\sigma} = \frac{1}{J}\mathbf{P}\mathbf{F}^\top = \frac{1}{J}\mathbf{F}\mathbf{S}\mathbf{F}^\top. \quad (4)$$

Note that $\mathbf{S} = \mathbf{S}^\top$ and $\boldsymbol{\sigma} = \boldsymbol{\sigma}^\top$ (balance of angular momentum), while $\mathbf{P}$ is in general unsymmetric, satisfying $\mathbf{P}\mathbf{F}^\top = \mathbf{F}\mathbf{P}^\top$.

2 Strong Form

2.1 Spatial (Eulerian / updated) form

Balance of linear momentum on the current configuration Ω_t reads: find $\mathbf{u}$ (equivalently the motion $\boldsymbol{\varphi}$) such that

$$\rho \dot{\mathbf{v}} = \text{div } \boldsymbol{\sigma} + \rho \mathbf{b} \quad \text{in } \Omega_t, \quad (5a)$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_t^D, \quad (5b)$$

$$\boldsymbol{\sigma} \mathbf{n} = \bar{\mathbf{t}} \quad \text{on } \Gamma_t^N, \quad (5c)$$

$$\mathbf{u}(\cdot, 0) = \mathbf{u}_0, \quad \mathbf{v}(\cdot, 0) = \mathbf{v}_0 \quad \text{in } \Omega_R, \quad (5d)$$

where ρ is the current mass density, $\mathbf{b}$ the body force per unit mass, $\mathbf{n}$ the outward unit normal on $\partial\Omega_t$, $\bar{\mathbf{t}}$ the prescribed traction per unit current area, and $\dot{\mathbf{v}} = \partial\mathbf{v}/\partial t + (\text{grad } \mathbf{v})\mathbf{v}$ the material acceleration. For quasi-statics the inertial term $\rho \dot{\mathbf{v}}$ is dropped. Angular momentum balance requires $\boldsymbol{\sigma} = \boldsymbol{\sigma}^\top$.

2.2 Referential (Lagrangian) form

Pulling (5) back to the reference configuration Ω_R using the Piola transform (4) and Nanson's formula $\mathbf{n} \, da = J\mathbf{F}^{-\top} \mathbf{N} \, dA$, the momentum balance becomes: find $\mathbf{u} : \Omega_R \times [0, T] \rightarrow \mathbb{R}^d$ such that

$$\rho_R \ddot{\mathbf{u}} = \text{Div } \mathbf{P} + \rho_R \mathbf{b} \quad \text{in } \Omega_R, \quad (6a)$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_R^D, \quad (6b)$$

$$\mathbf{P} \mathbf{N} = \bar{\mathbf{T}} \quad \text{on } \Gamma_R^N, \quad (6c)$$

$$\mathbf{u}(\cdot, 0) = \mathbf{u}_0, \quad \dot{\mathbf{u}}(\cdot, 0) = \mathbf{v}_0 \quad \text{in } \Omega_R, \quad (6d)$$

where $\rho_R = J\rho$ is the reference density, $\mathbf{N}$ the outward unit normal on $\partial\Omega_R$, and $\bar{\mathbf{T}} = \bar{\mathbf{t}}(da/dA)$ the prescribed nominal traction (force per unit *reference* area). All derivatives are taken with respect to $\mathbf{X}$ over the fixed domain Ω_R ; this is the natural setting for a total Lagrangian discretization.

The system is closed by a constitutive relation. For a hyperelastic material with stored energy density $\Psi(\mathbf{F})$ (per unit reference volume),

$$\mathbf{P} = \frac{\partial \Psi}{\partial \mathbf{F}}, \quad \text{or equivalently} \quad \mathbf{S} = 2 \frac{\partial \Psi}{\partial \mathbf{C}} = \frac{\partial \Psi}{\partial \mathbf{E}}. \quad (7)$$

3 Weak Form: Total Lagrangian Formulation

Define the trial and test spaces over the *fixed* reference domain,

$$\mathcal{S}_t = \left\{ \mathbf{u} \in [H^1(\Omega_R)]^d : \mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_R^D \right\}, \quad \mathcal{V} = \left\{ \mathbf{w} \in [H^1(\Omega_R)]^d : \mathbf{w} = \mathbf{0} \text{ on } \Gamma_R^D \right\}. \quad (8)$$

Taking the inner product of (6a) with $\mathbf{w} \in \mathcal{V}$, integrating over Ω_R , and using integration by parts together with $\mathbf{P} \mathbf{N} = \bar{\mathbf{T}}$ on Γ_R^N and $\mathbf{w} = \mathbf{0}$ on Γ_R^D , yields the total Lagrangian weak form:

Find $\mathbf{u}(\cdot, t) \in \mathcal{S}_t$ such that, for all $\mathbf{w} \in \mathcal{V}$,

$$\boxed{\int_{\Omega_R} \rho_R \ddot{\mathbf{u}} \cdot \mathbf{w} \, dV + \int_{\Omega_R} \mathbf{P}(\mathbf{F}) : \text{Grad } \mathbf{w} \, dV = \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV + \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA} \quad (9)$$

with $\mathbf{F} = \mathbf{I} + \text{Grad } \mathbf{u}$. Every integral is evaluated on the reference configuration, which never changes during the computation—the defining feature of the total Lagrangian formulation.

3.1 Equivalent form in the second Piola–Kirchhoff stress

Substituting $\mathbf{P} = \mathbf{F} \mathbf{S}$ and noting that $\mathbf{F} \mathbf{S} : \text{Grad } \mathbf{w} = \mathbf{S} : \mathbf{F}^\top \text{Grad } \mathbf{w} = \mathbf{S} : \delta \mathbf{E}[\mathbf{w}]$ (the last step using the symmetry of $\mathbf{S}$), where

$$\delta \mathbf{E}[\mathbf{w}] = \frac{1}{2} \left(\mathbf{F}^\top \text{Grad } \mathbf{w} + \text{Grad}^\top \mathbf{w} \mathbf{F} \right) \quad (10)$$

is the variation of the Green–Lagrange strain in the direction $\mathbf{w}$, the weak form (9) may equivalently be written

$$\int_{\Omega_R} \rho_R \ddot{\mathbf{u}} \cdot \mathbf{w} \, dV + \int_{\Omega_R} \mathbf{S}(\mathbf{E}) : \delta \mathbf{E}[\mathbf{w}] \, dV = \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV + \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA. \quad (11)$$

This is the form commonly used with $(\mathbf{S}, \mathbf{E})$ constitutive pairs (e.g. St. Venant–Kirchhoff or models expressed in terms of $\mathbf{C}$).

3.2 Remarks

- **Quasi-statics.** Dropping the inertial term in (9) gives the virtual work statement $\int_{\Omega_R} \mathbf{P} : \text{Grad } \mathbf{w} \, dV = \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV + \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA$, i.e. internal virtual work equals external virtual work.
- **Nonlinearity and linearization.** The weak form is nonlinear in $\mathbf{u}$ through $\mathbf{P}(\mathbf{F})$ (and geometrically through $\mathbf{F}$). A Newton scheme requires the consistent tangent $\mathbb{A} = \partial \mathbf{P} / \partial \mathbf{F} = \partial^2 \Psi / \partial \mathbf{F} \partial \mathbf{F}$, giving the directional derivative $D_{\Delta \mathbf{u}} \int_{\Omega_R} \mathbf{P} : \text{Grad } \mathbf{w} \, dV = \int_{\Omega_R} \text{Grad } \mathbf{w} : \mathbb{A} : \text{Grad } \Delta \mathbf{u} \, dV$; in the $(\mathbf{S}, \mathbf{E})$ form this splits into the familiar material and geometric stiffness contributions.
- **Follower loads.** A traction prescribed per unit *current* area (e.g. pressure $\bar{\mathbf{t}} = -p \mathbf{n}$) becomes configuration-dependent when pulled back: $\bar{\mathbf{T}} = -p J \mathbf{F}^{-\top} \mathbf{N}$, and then contributes to the tangent stiffness. The form above assumes $\bar{\mathbf{T}}$ is a dead (nominal) load; the follower pressure is treated in Section 4.
- **Relation to the spatial weak form.** Pushing (9) forward with $\mathbf{P} : \text{Grad } \mathbf{w} = J \boldsymbol{\sigma} : \text{grad } \mathbf{w}$ and $dv = J \, dV$ recovers the updated/spatial weak form $\int_{\Omega_t} \rho \dot{\mathbf{v}} \cdot \mathbf{w} \, dv + \int_{\Omega_t} \boldsymbol{\sigma} : \text{grad } \mathbf{w} \, dv = \int_{\Omega_t} \rho \mathbf{b} \cdot \mathbf{w} \, dv + \int_{\Gamma_t^N} \bar{\mathbf{t}} \cdot \mathbf{w} \, da$; the two statements are identical, differing only in the configuration on which integrals are evaluated.

4 Follower Pressure and Load Scheduling

4.1 Follower pressure

A pressure p acting per unit *current* area on Γ_R^p exerts the spatial traction $\bar{\mathbf{t}} = -p \mathbf{n}$. Nanson’s relation $\mathbf{n} \, da = J \mathbf{F}^{-\top} \mathbf{N} \, dA = \text{cof } \mathbf{F} \mathbf{N} \, dA$ pulls it back to the reference configuration,

$$\bar{\mathbf{T}}(\mathbf{u}) = -p J \mathbf{F}^{-\top} \mathbf{N} = -p \, \text{cof } \mathbf{F} \mathbf{N}, \quad (12)$$

so that the external virtual work of the pressure depends on the displacement. Moving it to the internal side, the quasi-static residual of (9) becomes

$$R(\mathbf{u}; \mathbf{w}) = \int_{\Omega_R} \mathbf{P}(\mathbf{F}) : \text{Grad } \mathbf{w} \, dV + \int_{\Gamma_R^p} p (\text{cof } \mathbf{F} \mathbf{N}) \cdot \mathbf{w} \, dA - \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV - \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA. \quad (13)$$

Its directional derivative in $\Delta \mathbf{u}$ adds the (in general non-symmetric) load stiffness

$$D_{\Delta \mathbf{u}} \int_{\Gamma_R^p} p (\text{cof } \mathbf{F} \mathbf{N}) \cdot \mathbf{w} \, dA = \int_{\Gamma_R^p} p [(\mathbf{F}^{-\top} : \text{Grad } \Delta \mathbf{u}) J \mathbf{F}^{-\top} \mathbf{N} - J \mathbf{F}^{-\top} (\text{Grad } \Delta \mathbf{u})^\top \mathbf{F}^{-\top} \mathbf{N}] \cdot \mathbf{w} \, dA, \quad (14)$$

using $DJ = J \mathbf{F}^{-\top} : \text{Grad } \Delta \mathbf{u}$ and $D\mathbf{F}^{-\top} = -\mathbf{F}^{-\top}(\text{Grad } \Delta \mathbf{u})^\top \mathbf{F}^{-\top}$. Because the tangent is not symmetric, the linear solver must be a general Krylov method (GMRES); the conjugate gradient option is refused for inputs with a follower pressure.

In the code (`kernels/follower_pressure.hpp`) the term is a boundary face integrator of the nonlinear form: on each boundary face of Γ_R^p the displacement gradient of the adjacent element gives $\mathbf{F}$ at the face quadrature points, $\mathbf{N} dA$ is the area-weighted reference normal of the face map, and the tangent (14) is obtained by seeding $\mathbf{F}$ with dual numbers in the direction of each element displacement degree of freedom, exactly as the material tangent is. The same face kernel serves the displacement block of the mixed u - p formulation. The dead normal pressure, $\bar{\mathbf{T}} = -p \mathbf{N}$ per unit *reference* area, remains a linear-form term; the two coincide at $\mathbf{F} = \mathbf{I}$ and differ at $O(p^2)$ for small loads. Verification: the tangent against finite differences on distorted quadrilateral and hexahedral meshes, and the inflation of a thick-walled incompressible cylinder against the closed form $P = \mu [\ln(B/A) - \ln(b/a) + \frac{1}{2}(A^2 - a^2)(b^{-2} - a^{-2})]$, $b^2 = B^2 - A^2 + a^2$ (plane strain, neo-Hookean).

4.2 Load scheduling

Quasi-static loading is parametrised by a pseudo-time $t \in [0, 1]$. Every prescribed displacement, traction, pressure and body force i carries its own scalar schedule $s_i(t)$ (a ramp over $[t_0, t_1]$, a constant, or a piecewise-linear table) and enters the residual as $s_i(t)$ times its data; data may in addition depend on t through an expression $f(\mathbf{X}, t)$. The default ramp $s_i = t$ for every entry recovers the proportional load path, in which $t < 1$ solves a scaled problem and $t = 1$ the problem of the weak form. The stepper advances t through user-defined breakpoints and, when a Newton solve fails, restores the last converged state and halves the increment (bisection), returning to the planned breakpoint afterwards. Because the hyperelastic problem without follower loads is conservative, the converged state at $t = 1$ does not depend on the path; staged loading (one load ramped and held, a second one ramped afterwards) is therefore a test of the implementation rather than a different problem. Follower pressures are not conservative, and their solutions can be path dependent in the presence of instabilities.